

SOP: Dataset Profiling and Baseline Modeling

This SOP defines a simple workflow for scientific dataset analysis. First, upload the CSV file into the active project workspace. The upload should persist the dataset immediately so the agent can inspect it without requiring a separate save step.

Second, profile the dataset. The system should report row count, column names, numeric and categorical fields, missing values, and candidate target variables. For regression tests, `yield_pct` or `conversion_pct` are valid targets. For classification tests, `quality_pass` is a valid target.

Third, train a baseline model only after a target column is confirmed. If the user says train a model without specifying a target, the assistant should ask which target column to use. It should remember the pending training action and continue when the user replies with the target.

Fourth, persist every model run in the workspace. Chat should summarize the result, but the model run should also appear in the project model history with metrics and feature importance.

This SOP is intentionally small so RAG retrieval should find the sections about target variables, missing values, and model persistence.